CASE STUDY 14

Warehouse Inventory Reconciliation

1. PROBLEM STATEMENT

Compare opening stock, incoming quantities, dispatched quantities, and recorded closing stock for several products.

Detect discrepancies and calculate replenishment needs.

2. OBJECTIVES

• Calculate expected closing stock.

• Compare expected and recorded stock.

• Identify products that need replenishment.

3. ASSUMPTIONS

• Each product record contains name, opening stock, incoming stock, dispatched stock, and recorded closing stock.

• Expected closing stock = opening + incoming – dispatched.

• Target stock for each product is 20 units.

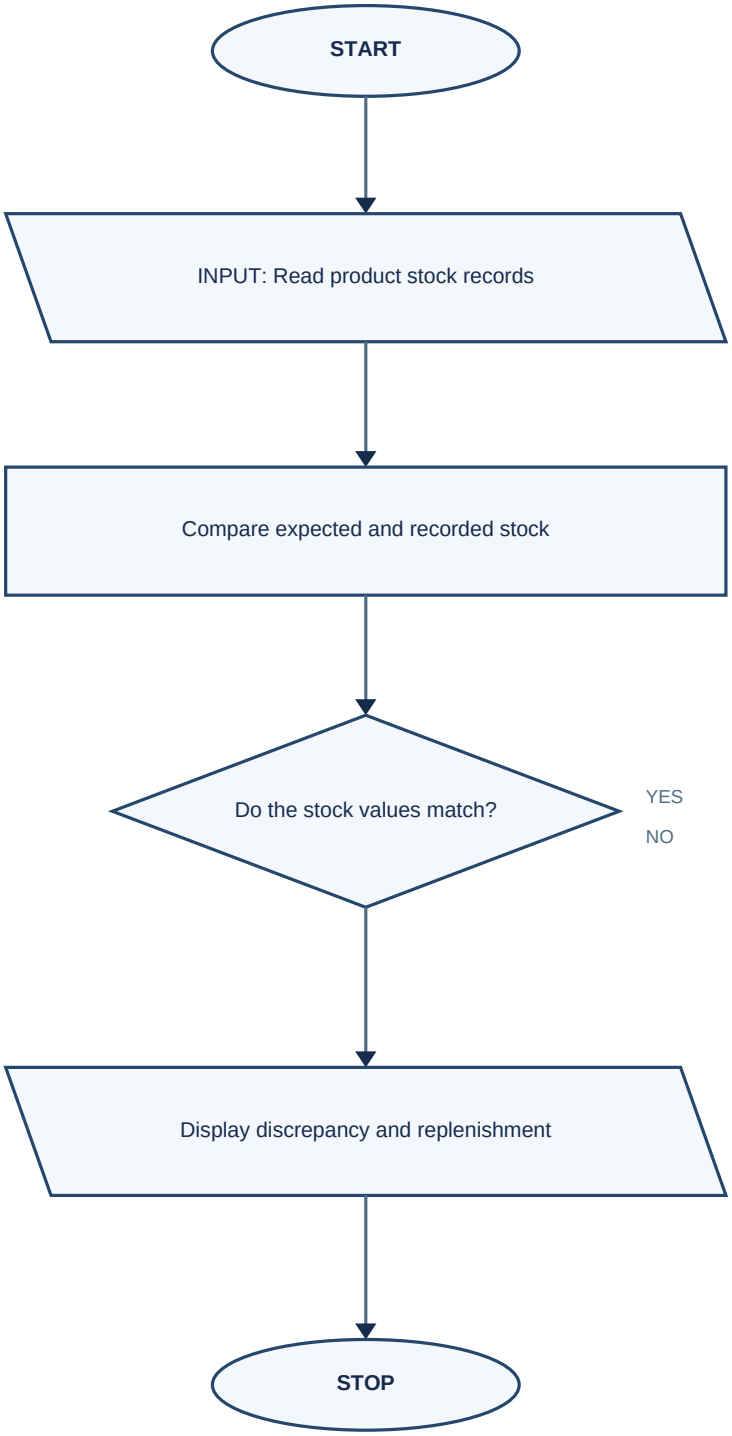
• Negative stock quantities are invalid.

4. ALGORITHM

1. Read each product's opening, incoming, dispatched and recorded closing stock.
2. Calculate expected closing stock as opening plus incoming minus dispatched.
3. Compare expected closing stock with recorded closing stock and report the status.
4. Calculate replenishment as the difference from target stock when expected stock is below 20; otherwise use zero.
5. Display expected stock, recorded stock, discrepancy status and replenishment.

5. FLOWCHART

Oval: START/STOP | Parallelogram: input/output | Rectangle: process/calculation | Diamond: decision



6. PSEUDOCODE

```
BEGIN
INPUT product stock records
FOR each product
expected = opening + incoming - dispatched
IF expected != recorded: report discrepancy
ELSE: report matched
IF expected < target_stock
needed = target_stock - expected
ELSE: needed = 0
DISPLAY expected, recorded and replenishment
END
```

7. PYTHON SOURCE CODE

```
# name, opening, incoming, dispatched, recorded closing
products = [
["Pen", 50, 30, 40, 40],
["Book", 20, 10, 15, 18],
["File", 25, 5, 20, 10]
]
target_stock = 20
print("WAREHOUSE STOCK REPORT")
for p in products:
name, opening, incoming, dispatched, recorded = p
expected = opening + incoming - dispatched
print("\\nProduct:", name)
print("Expected closing stock:", expected)
print("Recorded closing stock:", recorded)
if expected != recorded:
print("Status: DISCREPANCY FOUND")
else:
print("Status: MATCHED")
if expected < target_stock:
needed = target_stock - expected
else:
needed = 0
print("Replenishment required:", needed)
```

8. TEST CASES & EXPECTED RESULTS

TEST INPUT / CONDITION REASONING EXPECTED RESULT

Pen: 50 + 30 – 40 Expected 40; recorded 40 Matched; replenishment 0

Book: 20 + 10 – 15 Expected 15; recorded 18 Discrepancy 3; replenishment 5

File: 25 + 5 – 20 Expected 10; recorded 10 Matched; replenishment 10

9. SAMPLE OUTPUT

```
WAREHOUSE STOCK REPORT
Product: Pen
Expected closing stock: 40
Recorded closing stock: 40
Status: MATCHED
Replenishment required: 0
Product: Book
Expected closing stock: 15
Recorded closing stock: 18
Status: DISCREPANCY FOUND
Replenishment required: 5
Product: File
Expected closing stock: 10
Recorded closing stock: 10
Status: MATCHED
Replenishment required: 10
```

10. CONCLUSION

The inventory program detects differences between calculated and recorded stock and identifies products that need replenishment.

SYLLABUS CONCEPTS Variables • Data Types • Operators • Conditions • Loops • Lists • Nested Lists • Tuples • String Operations